F1 Forecast

A Data-Driven Race Outcome and Tyre Strategy Prediction System

Project Synopsis

1. Introduction

Formula 1 generates enormous volumes of telemetry, timing, and strategic data every race weekend, yet most publicly available prediction tools rely on superficial statistics (past wins, grid position) rather than the underlying physical and strategic factors that determine race outcomes — tyre degradation, weather shifts, pit strategy, and driver form. F1 Forecast is a Python-based analytics system that uses official, publicly available F1 datasets to model these deeper factors and generate explainable race predictions.

2. Problem Statement

Existing F1 prediction content (fan blogs, basic machine learning tutorials) typically uses shallow historical statistics and produces a single opaque prediction with no reasoning behind it. There is no accessible tool that combines tyre degradation modelling, race outcome prediction, and explainability (why the model predicts what it predicts) into one coherent, reproducible pipeline built on official data sources.

3. Objectives

- Build an automated pipeline to ingest and clean official F1 timing, weather, and tyre data.
- Model tyre degradation per compound and circuit.
- Predict race outcomes (finishing positions/points) using engineered features and gradient boosting.
- Provide explainability for predictions using SHAP, so outputs are interpretable rather than a black box.
- Enable counterfactual analysis (e.g. “what if the driver pitted a lap earlier?”).
- Present results through an interactive dashboard.

4. Scope

The project covers race outcome and tyre-strategy prediction for Formula 1 races using publicly available historical and live session data (via the FastF1 and Jolpica-F1 APIs). It does not cover other motorsport categories, real-time in-race betting/odds systems, or live broadcast integration. The system is intended for analysis and demonstration purposes rather than official team strategy use, given the depth of proprietary data real teams have access to that is not publicly available.

5. Proposed Solution / Methodology

The project follows a modular pipeline consisting of the following stages:

- **Data Ingestion** — pull lap times, tyre compounds, weather, and session data via FastF1 and Jolpica-F1, cached locally for reproducibility.
- **Feature Engineering** — construct features such as recent driver form, circuit characteristics, tyre compound behaviour, and weather conditions.
- **Tyre Degradation Modelling** — a regression model estimating performance drop-off per lap, per compound, per circuit.
- **Race Outcome Modelling** — an XGBoost-based model predicting finishing position or points using engineered features.
- **Explainability Layer** — SHAP values to show which features drove each prediction, plus counterfactual scenario testing (e.g. altering pit lap or tyre choice).
- **Presentation Layer** — a Streamlit dashboard visualising predictions, degradation curves, and explainability outputs.
- **Validation** — backtesting against actual historical race results to measure accuracy.

6. Features / Modules

- Data Ingestion Module
- Feature Engineering Module
- Tyre Degradation Model
- Race Outcome Prediction Model
- Explainability & Counterfactual Engine
- Interactive Dashboard
- Backtesting & Validation Module

7. Technology Stack

Category	Technology
Language	Python
Data Sources / APIs	FastF1, Jolpica-F1
Libraries	pandas, NumPy, scikit-learn, XGBoost, SHAP
Dashboard	Streamlit
Version Control	Git & GitHub
Environment	VS Code, Python venv

8. System Requirements

- **Hardware:** Standard laptop/PC, minimum 8GB RAM recommended for model training.
- **Software:** Python 3.11+, VS Code, Git.
- **Browser:** Any modern browser (for the Streamlit dashboard).
- **Operating System:** Windows / macOS / Linux (cross-platform).

9. Expected Outcomes

A working, reproducible pipeline that ingests official F1 data and outputs interpretable predictions for race outcomes and tyre strategy, presented through an interactive dashboard. Benefits include a transparent, explainable alternative to black-box prediction tools, and a portfolio-grade demonstration of applied data science and machine learning engineering.

10. Future Scope

- Live session integration for in-race prediction updates.
- Extension to qualifying performance prediction.
- Driver-vs-driver head-to-head simulation mode.
- Deployment as a publicly hosted web application.

11. Conclusion

F1 Forecast demonstrates how official, publicly available motorsport data can be transformed into a transparent, explainable prediction system — bridging the gap between simplistic fan-facing predictors and the opaque proprietary models used by professional teams, while serving as a comprehensive applied project in data pipelines, machine learning, and interpretability.